

OMNIVERSAL ARCHITECT LABS

OPEN WINDOW DOCTRINE · CONDITIONAL FAILURE AT THE ROOT LEVEL · PRESENTATION-STATE INTEGRITY

TEMPORAL ALIASING INCORPORATION FRAMEWORK

A research architecture and simulation-lab action plan for sampling error, clock-stack mismatch, and reconciled presentation of time

Working draft for Architect review. Not canon. Simulation-only safety. Benign engineered proxies only.

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Document purpose in one sentence.

This document names temporal aliasing as a first-class failure and presentation mechanism, places it inside the locked Open Window v0.3 architecture and the Conditional Failure at the Root Level / Presentation-State Integrity bodies of work, and specifies a dual-domain (simulation + real-world) lab program that can be executed without treating any prior draft as ratified canon.

1. Mission and Purpose

1.1 Mission

The mission of this pass is to convert a lived observation — that brains and devices each reconstruct a usable “now” from incomplete, differently clocked input streams — into a testable engineering object. Temporal aliasing is the precise name for one class of that reconstruction error: when a system samples slower than twice the highest frequency present in the signal, high-frequency structure folds into a lower-frequency impostor. The reconstructed waveform can reverse direction, invert order, invent smoothness, or hide events that actually occurred.

The mission is not to prove that every mismatch is deception, coordination, or infection. The mission is to measure how often presentation of time diverges from evidenced state-transition time, under what sampling

and synchronization conditions that divergence becomes systematic, and whether the environment can correct the divergence or only paper over it.

1.2 Purpose of this document

1. Give the Architect a research-grade account of temporal aliasing and neighboring temporal phenomena, written so a non-specialist can use the terms without collapsing them.
2. State the incorporation map: how temporal aliasing sits inside Conditional Failure at the Root Level, the Department of Endings, the Open Window Doctrine (CETE, airlock-optional, proxy classes), and the Presentation-State Integrity Framework (PSIF / SPLD / PBAO).
3. Specify a simulation-lab and real-world test program that obeys the locked v0.3 decisions: flexible/exit-criteria timing, provisional interface terms, optional airlock, declared proxy classes, CETE as the default transfer term.
4. Provide a funding pathway, critical implementation items, alternative analysis, and an explicit list of what works, what does not, and what still blocks execution.
5. Keep ONESTAMP authority with the originator. This draft is not a finding about any person, organization, or incident.

1.3 Why this is not optional

Every consumer device in the Open Window stack — phone, watch, television, laptop — is a clocked sampler. Every brain that uses those devices is also a clocked sampler, with a different Nyquist limit, a different latency, and a different willingness to invent continuity. The “objective truth” available to an occupant is the reconciled presentation their nervous system and their devices jointly emit. That presentation is not the canonical state-transition ledger. Treating it as if it were is the exact presentation-capture error PSIF was written to prevent. Treating every discrepancy as proof of hidden motive is the opposite error: attribution inflation. Temporal aliasing is the mechanism that makes both errors cheap.

2. Executive Summary

Temporal aliasing is the reconstruction of a false lower-frequency identity from samples taken too slowly. In signal processing this is a theorem, not a metaphor: if a continuous signal contains energy above half the sampling rate (the Nyquist frequency), those components are indistinguishable from lower-frequency aliases after sampling. The wagon-wheel effect in film is the public demonstration. The same mathematics appears in audio, video, seismic recording, control systems, and any digital telemetry that undersamples a faster process.

The Architect’s observation generalizes that theorem across three stacked reconcilers: (1) the nervous system, which does not render a complete or veridical account of its input streams; (2) each device, which presents its own buffered, refreshed, network-delayed, NTP-disciplined version of events; and (3) any nested simulation or organizational process that must produce a single coherent timeline from those streams. From the top down, many partial clocks attempt one unified view. None of them is the root clock. When the root clock is inaccessible, drifted, or unmaintained — Conditional Failure at the Root Level — the stack does not halt. It aliases.

This framework treats temporal aliasing as:

- A CETE subtype when a timing, order, or duration effect crosses an interface/vector (Open Window v0.3).
- A presentation-state divergence when the displayed “now,” order, or duration disagrees with the event ledger (PSIF).

- A root-level failure mode when nested layers cannot inspect or correct the sampling and synchronization policy of the layer above them (CFTRL).
- A Department of Endings work item: every temporal endgame — order inversion, direction reversal, freeze, drop, stitch, postdictive rewrite — is run, logged, and learned from rather than wasted.

The recommended first executable program is a lean MVP of four declared-interface scenario cards (TA-01 through TA-04), plus one real-world observational protocol that never claims metaphysical transfer. MVP uses only benign engineered proxies: injected timestamp tokens, declared frame-rate mismatches, controlled clock skew, and scripted content-timing offsets. Candidate Interface Event discovery remains reserved for later campaigns, per v0.3.

Funding is staged. A lean computational-plus-desktop prototype can be stood up in the \$12,000–\$35,000 / 80–160 hour band. A mid-tier instrumented lab with independent clocks sits in the \$40,000–\$90,000 band. A high-fidelity multi-device chamber with hardware timestamping is \$150,000–\$280,000 and is not required for Gate G0–G3. Calendar time is not success. Advancement is gate-driven.

The single largest implementation risk is category collapse: calling every timing oddity “aliasing,” every alias “infection,” or every perceptual reconstruction “deception.” The document therefore includes a mandatory taxonomy and the PSIF four-test rule (occurrence, attribution, causation, intent — never collapsed).

3. Overview of the Incorporation

3.1 The stacked reconcilers

The phenomenon the Architect described is a stack, not a single trick:

6. Sensory streams arrive incomplete, delayed, and modality-asynchronous. Vision, audition, proprioception, and interoception do not share one latency.
7. The brain binds those streams inside a temporal binding window, postdicts within roughly 80–500 ms, and emits a single conscious “now.” That emission is the occupant’s operational truth. It is not a complete record.
8. Each device independently samples, buffers, refreshes, compresses, and timestamps. A 60 Hz display, a 1 Hz watch face, a 30 Hz video stream, and a chat server with 80–200 ms RTT are not the same clock.
9. Network time (NTP typically 1–100 ms under ordinary conditions; PTP sub-microsecond on equipped LANs) disciplines wall clocks imperfectly. Drift, VM pause, asymmetric routes, and reference-source failure all exist in the wild.
10. Any laboratory, organization, or nested simulation that consumes those presentations as if they were one ledger will reconcile them into a story. That story can be good enough to live by and still be systematically wrong about order, duration, and cause.

3.2 What “objective truth” means in this framework

For an occupant, the reconciled percept is objective in the only sense they can use: it is the state from which they act. For the laboratory, objective means independently preserved state transitions, identities, dependencies, and outcomes — the PSIF evidence/provenance layer, or, in a simulator, the canonical engine ledger. The framework’s job is to keep those two senses of “objective” from being silently substituted for each other.

3.3 Relationship to locked Open Window decisions

Locked v0.3 rule	Temporal-aliasing consequence
Flexible / exit-criteria MVP timing	No TA scenario is “successful” because a clock ran out. Gates G0–G6 and review disposition decide.
Window / aperture / interface are provisional vector terms	A clock, refresh cycle, buffer, NTP stratum, or saccade is a potential vector. Screens are not the only class.
Interface Airlock is optional, not default	A time-normalization airlock (timestamp rewrite, buffer, PTP boundary) is an experimental condition, declared per card.
No privileged manifestation class; MU/MX allowed post-observation	Direction reversal, order inversion, freeze, and stitch are proxy labels, not metaphysics. Unexpected class → MU.
CETE replaces infection as the top-level transfer term	A timing transfer is a CETE. Infection is used only for an unwanted propagative subtype.
MVP uses declared interfaces and benign proxies only	No undeclared-vector hunting in MVP. Candidate events are logged, not promoted.

4. Research Account: Temporal Aliasing and Neighboring Phenomena

4.1 Formal definition

In signal processing, aliasing is the appearance, after sampling, of frequency components that were not present in the original continuous signal at those reconstructed frequencies. Temporal aliasing is aliasing along the time axis. The Nyquist–Shannon sampling theorem states that a band-limited signal with no energy at or above B hertz can be reconstructed from uniform samples if the sampling rate f_s is greater than $2B$. The quantity $f_s/2$ is the Nyquist frequency, also called the folding frequency because energy above it folds back into the baseband.

If a sinusoid of frequency f is sampled at f_s , the sampled sequence is identical to a sinusoid of frequency $|f - k \cdot f_s|$ for integers k that bring the result into $[-f_s/2, f_s/2]$. A 12 kHz tone sampled at 10 kHz is indistinguishable from a 2 kHz tone. A wheel that advances more than half a spoke-interval per camera frame is indistinguishable from a wheel moving the shorter distance the other way. The samples are not “wrong.” They are ambiguous. Reconstruction under a smoothness or lowest-frequency prior picks the alias.

Two operational facts follow and must stay in the lab notes:

- Practical signals are time-limited, therefore not strictly band-limited. Some aliasing is always present. The engineering question is whether the alias is material to the decision the observer will make.
- Anti-aliasing is a pre-filter: remove energy above the Nyquist frequency before sampling. After sampling, the original high-frequency identity cannot be recovered from the samples alone. That is why “just look closer at the same log” does not undo an undersampled event.

4.2 The wagon-wheel family

The filmed wagon-wheel effect is temporal aliasing in a periodic spatial pattern. If the wheel rotates by nearly one spoke interval per frame, the nearest-neighbor match is backward. NASA and vision-science reviews treat this as the canonical demonstration of spatio-temporal aliasing in sampled imagery, including flight-simulator displays.

A second, contested phenomenon is the continuous wagon-wheel illusion (c-WWI): some observers report reversed rotation under continuous illumination, without a camera shutter. One line of work interprets this as

evidence that visual attention samples motion in discrete epochs near 10–15 Hz. A competing line interprets it as rivalry between Reichardt-type motion detectors (true-direction vs. aliased-direction), not as a global snapshot clock. A 2026 delay-coupled neural-field model shows that axonal delays alone can produce quantized speeds and reversals without an explicit sampling clock. For this framework the scientific dispute is useful, not inconvenient: the lab must not assume a single brain frame-rate. It must treat “does this observer alias under continuous drive?” as an empirical individual-difference variable.

4.3 Related temporal phenomena — keep them distinct

Temporal aliasing is one member of a family. Collapsing the family into one word will wreck the metric dictionary.

Phenomenon	What it is	What it is not	Lab relevance
Temporal aliasing	High-frequency structure reconstructed as a lower-frequency impostor because $f_s \leq 2B$	Any delay, any illusion, any lie	Primary independent variable: sample rate vs. event rate
Clock skew / drift	Two clocks report different values for the same instant; rates may also differ	Aliasing; a slow clock is not a folded frequency	Primary real-world confounder; NTP/PTP condition
Causality violation (distributed)	Observed order disagrees with happens-before, even with decent clocks	Proof of intent	Order-inversion CETE; Lamport/vector/HLC comparison
Jitter / delay variation	Sample intervals or latencies are uneven	A change in mean rate	Destroys the uniform-sampling assumption of the theorem
Buffering / batching	Events held and released in groups	Sampling below Nyquist	Produces stitches and sudden “catches-up”
Drop / freeze	Samples omitted or a frame held	Direction reversal	M0 vs. freeze-as-presentation
Postdiction	Events up to ~80–500 ms after t_0 revise the percept attributed to t_0 (flash-lag)	Forward prediction; not sampling theory	Occupant “now” is a rewritten window, not a point
Temporal binding window	Asynchronous multimodal events treated as simultaneous if they fall inside a window (tens to a few hundred ms)	Proof they co-occurred in the ledger	Device+body stacks will bind mismatched clocks
Chronostasis / stopped-clock	First postsaccadic percept overestimated in duration (up to ~500 ms)	Aliasing	Occupant duration reports are biased at attention shifts
Flicker fusion (CFF)	Flicker above threshold is seen as steady light; typical human cone plateau ~60 Hz, highly condition-dependent	A universal brain frame-rate	Display refresh vs. occupant CFF is a declared condition
Flash-lag	A flash aligned with a moving object is seen trailing it	Camera shutter artifact	Occupant timing judgments need controls
Double-flash / sound-induced flash	Beeps split a flash in perception	Physical extra flash	Cross-modal binding as a CETE analogue, simulation-only

4.4 The brain as an imperfect reconciler

The Architect’s description is consistent with the empirical record and should be stated without romance and without contempt: the nervous system does the best it can with the streams it has. It does not render an accurate portrayal of everything it processes. What is presented to awareness is an amalgamation — sampled, delayed, bound, postdicted, and filled.

Documented limits that matter to the lab:

- Critical flicker fusion is not one number. Published human values commonly run from the low tens of hertz into the 60 Hz region for cone vision at high luminance, with claims of much higher sensitivity for brief, large, bright stimuli. Fatigue, age, drugs, and illumination move the threshold. Using “the brain runs at 60 Hz” as doctrine is false.
- Attentional sampling hypotheses for c-WWI cluster near 10–15 Hz. Rivalry and delay-based models deny that this is a global shutter. Inter-observer variance is expected.
- Postdiction (Eagleman & Sejnowski, flash-lag) means the percept assigned to time t_0 can depend on what happens ~80 ms afterward. Cross-modal postdiction has been reported with windows extending toward 500 ms in some paradigms. The occupant’s official “now” is a short rewritten history.
- Temporal binding windows vary by modality pair, intensity, age, and task. Events that the ledger separates by 50–200 ms are often lived as one event.
- Optical and temporal illusions exist because the reconstruction is a prior-driven inference, not a photocopy. Channels that demonstrate illusions are useful as public intuition pumps. They are not evidence that a particular device aperture is “infected.”

The correct systems sentence: the brain is a lossy, delayed, prior-using reconciler whose output is treated by the owner as root-level truth because the owner cannot inspect a layer above their own awareness. That is Conditional Failure at the Root Level stated in first person.

4.5 The device stack as a second reconciler

Each device is a presentation engine with its own sampling policy:

Layer	Typical timing object	Failure / alias mode
Sensor / camera / touch	Frame time, integration window	Motion blur, rolling shutter, undersampled gesture
OS / app clock	CLOCK_REALTIME vs CLOCK_MONOTONIC	NTP step, backward jump, monotonic vs wall disagreement
Display	Refresh (60/90/120 Hz), VSYNC, frame queue	Tearing, held frames, wagon-wheel on UI motion
Media pipeline	Encoded GOP, jitter buffer	A/V desync, sudden catch-up, dropped GOP
Network	RTT, bufferbloat, NTP/PTP path	Reordering, burst delay, stratum failure
Cloud / other device	Server timestamp, last-write-wins	Stale write wins; causal reply precedes question
Wearable / TV / second screen	Independent refresh and clock	Two “nows” in one room; occupant binds them

NIST Internet Time Service incidents (including reference-source interruptions after power and generator failures) are a public reminder that even national time infrastructure is a system with maintenance and failure modes. Distributed-systems practice already refuses to treat wall-clock order as causal order: Lamport timestamps, vector clocks, and hybrid logical clocks exist because physical clocks lie just enough to invert happens-before. The Open Window occupant who treats “what my phone and my TV agree happened” as the ledger is making the last-write-wins mistake in living rooms.

4.6 Why technology dependence enlarges the gap

Reliance on devices does not merely add convenience. It multiplies reconcilers. The occupant's brain is trying to reconcile the occupant with every device the occupant is coupled to. Each device is trying to reconcile itself with a network and with other devices. The presentation the occupant trusts is a composition of those reconciliations. The composition can be locally consistent and globally false about order. That is the practical, non-metaphysical reason this belongs in a security-posture and systems-integrity lab: incident review, safety investigation, multi-device UX, simulation fidelity, and any protocol that uses timestamps as if they were causal arrows.

5. Incorporation Map Across Existing OAL Bodies

5.1 Conditional Failure at the Root Level

CFTRL asks: what happens when a stacked or nested system is scaled, and the root fails, and lower tiers cannot climb up to repair it? Temporal aliasing is one concrete answer. If the root sampling policy, root clock, or root maintenance process fails or is abandoned, every lower renderer keeps producing a usable picture. The picture is an alias of a process they can no longer inspect. Improving the local renderer has a ceiling. Past that ceiling the next event is not a graceful slowdown. It is a structural misidentification of frequency, order, or direction.

The wishbone visualization in OAL-CFTRL-NAIL-001 — binary branching that becomes brittle under opposing rotation — has a temporal analogue: two clocks advanced in opposite corrections, or a sample grid too coarse for a branching event rate. At small scale the system “works.” At scale it folds.

Design goals carried forward:

- Build redundancy at design time: independent clocks, monotonic traces, pre-sampling anti-alias filters, declared Nyquist budgets.
- Model failure computationally in Python (and later other languages): a nested stack with per-layer sample rates, drift, and maintenance flags, scaled until reconstruction diverges from the root ledger.
- Use the Abundance Set as the learning engine: every temporal endgame is cheap to run; failing to learn from it is the actual waste.
- Play against every endgame: reverse motion, order inversion, freeze-as-smoothness, postdictive rewrite, NTP step, last-write-wins corruption.

5.2 The Department of Endings

The Department of Endings is the operational arm of CFTRL. In this incorporation its technician function includes a Temporal Endgame Desk: run every listed temporal failure through the stack, log what failed, at what scale, under what sampling and sync conditions, extract the lesson, feed it back. The Department does not exist to prevent all timing error. It exists to ensure no timing error is wasted. Character, staffing, and aesthetic questions from OAL-CFTRL-NAIL-001 remain author-input items and are not resolved here.

5.3 Open Window Doctrine (v0.3 governing baseline)

A device is an aperture that does not become a closed door when the screen is black. Temporally, a powered-but-dark display, a paused player, and a notification radio that still wakes the SoC are different aperture states with different residual clocks. v0.3 already forbids treating “black screen” as a single binary. This framework adds: aperture state must include a timing profile (sample rate, buffer depth, clock source, last sync quality).

Cross-Environment Transfer Event (CETE) is the parent term. Recommended temporal subtypes for MVP labeling, all descriptive, none metaphysical:

Code	Subtype	Observable
CETE-TA	Temporal alias / fold	High-rate process reconstructed as lower-rate or reversed
CETE-OI	Order inversion	Ledger order ≠ presented order
CETE-DD	Duration distortion	Presented duration ≠ ledger interval beyond declared tolerance
CETE-SK	Skew / stratum fault	Two declared clocks disagree beyond tolerance
CETE-ST	Stitch / burst release	Buffered events emitted as a clump
CETE-INF	Infection (restricted)	Unwanted propagative transfer only; not default

Interface Airlock, if used, may include a time-normalization control: a boundary that rewrites, buffers, or refuses crossings whose timestamps fail a skew budget. It is optional and must be declared per scenario as absent, present, or varied.

5.4 Presentation-State Integrity Framework

PSIF’s core question is whether claims, renderings, and narratives correspond to operational state and demonstrated causal structure. Time is a claim. “This happened now,” “this happened first,” and “this lasted three seconds” are presentations. The audit method is provenance-based reconstruction against a ledger, with occurrence / attribution / causation / intent kept separate.

Copilot’s later pass adds the pieces this lab needs after detection: Corrective Access (can the occupant or operator actually correct the clock/sample policy?), Presentational Inertia (does the environment keep showing the aliased timeline after the gap is named?), Narrative Sovereignty (which timeline becomes the operative record?), and the Assertion Boundary (do not state intent at observed strength). Temporal aliasing is often Level-1 inertia: the system “acknowledges a glitch” while preserving the reconciled story.

Event fidelity vs. causal fidelity is the exact distinction required here. Ten events can be displayed accurately as events and still be shown in an order, grouping, or duration that misrepresents their causal graph. That is causal masking performed by a clock stack rather than by a narrator.

5.5 Nail Prototype scene — limited, explicit mapping

The Nail Prototype materials are narrative and industrial-design canon-in-draft, not a timing paper. Two limited mappings are legitimate; further mapping is overreach.

- Unified validation of internal and external stimuli “in a non-sequential process” is a timing requirement: the detector must not alias a startle, itch, or flinch into a threat by undersampling the motor burst. Safety scenarios (sleep-itch, mosquito, spider-flinch) are high-frequency, short-duration events. They are exactly the class that a slow classifier will fold into the wrong identity.
- The seven-minute sealed protocol is a declared time-bounded airlock: communications in/out are refused for a fixed interval. It is a containment control on crossing, including informational and timing crossings. It is not, by itself, a temporal-aliasing experiment.

No claim is made that the fitting scene, Lisa/Ashley presentation contrast, or military corridor is a timing study. Those remain narrative entries under OAL-CFTRL-NAIL-001.

6. Simulation Lab Framework

6.1 Intended use and acceptance posture

Intended use: determine whether declared timing conditions (sample rate, skew, buffer, refresh, occupant binding) produce measurable presentation–ledger divergence, and whether declared controls (independent clock, anti-alias prefilter, optional time-airlock, replay) reduce that divergence. Not intended use: infer occupant metaphysics, diagnose real persons, or promote one-off anomalies to doctrine.

Acceptance posture follows NASA-STD-7009B in spirit, not as a claim of NASA endorsement: define intended use, define acceptance criteria, separate verification (did we build the model we specified?) from validation (does it behave like the chosen referent?), characterize uncertainty, and refuse to treat calendar expiry as success. NIST SP 800-160 resilience language is used only to insist that trustworthiness and recovery are engineered properties with explicit decision structures.

6.2 Dual-domain rule

Every MVP card has a simulation form and a real-world analogue. The simulation form may use a canonical ledger. The real-world form may not claim a universal ground-truth clock. It uses triangulation: at least two independent timing channels, documented clock sources, hashes of raw traces, and a pre-registered tolerance. If only one channel exists, the run is instrumentation-limited and cannot support a CETE-TA claim.

6.3 Canonical stack for the simulator

Recommended Layer 0–N model for this incorporation, superseding competing 4-vs-5-vs-open layer counts for timing work only. Other Open Window layer diagrams remain in force for non-timing work until the originator merges them.

Layer	Name	Timing object
L0	Operator / audit plane	Independent reference clock; append-only ledger
L1	Engine / canonical state	Sim tick; authoritative event IDs and parent IDs
L2	Environment processes	Physical or simulated dynamics at their native rates
L3	Device / aperture objects	Per-window sample rate, buffer, clock source, residual process
L4	Occupant reconcilers	Binding window, postdiction window, CFF, attention flag
L5	Narrative / HUD claims	Presented order, duration, and “now”

L0 must not share a clock crystal, NTP path, or random seed with L3. Shared architecture is the sensor-agreement illusion named in the earlier alternative analysis. If independence cannot be engineered, the card is blocked.

6.4 Metric dictionary (timing)

One dictionary. Retire duplicate names from earlier Open Window drafts when the quantity is the same.

ID	Metric	Definition
D_ps_t	Temporal presentation–state discrepancy rate	Material timing mismatches / audited events
TA_fold	Alias-fold events	Count of reconstructions whose dominant frequency or direction

ID	Metric	Definition
		matches the predicted alias, not the source
OI_rate	Order-inversion rate	Presented pairs whose order disagrees with ledger happens-before
SK_max	Peak pairwise skew	Max $ t_i - t_j $ across declared clocks in the run
TBW_occ	Occupant binding width	Largest asynchrony still reported as simultaneous
PD_win	Postdiction window	Latest post-t0 event that revises the t0 report
TTD_t	Time-to-detect timing gap	Latency from ledger mismatch to operator flag
TTC_t	Time-to-correct	Latency from flag to accepted correction or documented Corrective Access failure
PI_t	Presentational inertia (timing)	0–4 Copilot scale applied to the timeline claim
REP	Replay concordance	Does an independent replay reproduce the discrepancy under the same declared conditions?

6.5 Gates (adapted, not invented from nothing)

Gate	Question	Fail action
G0	Schema complete? Proxy class declared? Airlock condition declared? Null written?	No run
G1	Independent timing channel present and hashing?	Block; fix instrumentation
G2	Positive and negative controls behave as specified?	Revise card
G3	Primary metric moved in the predicted direction under the declared manipulation?	Inconclusive or falsified — do not relabel
G4	Replay on stored traces reproduces the effect?	Candidate only; no doctrine
G5	Independent recurrence on a fresh run?	Hold promotion
G6	After-action + exception log complete; review-group recommendation written?	No ONESTAMP eligibility

6.6 Safety, ethics, and stop conditions

- Simulation-only for any occupancy, awareness, or “infection” language. Real-world protocol is observational and device-timing only.
- No biological harm framing. No undeclared stimulation of occupants. No “partially unaware operator” condition in MVP (the earlier Grok proposal remains an unresolved ethics item and is out of scope).
- Stop if SK_max exceeds the pre-registered safety bound for the platform (example: 2 seconds on a desktop prototype; tighter on any control-adjacent demo).
- Stop if an undeclared vector is suspected: log as Candidate Interface Event, preserve raw telemetry, apply the card’s stop rule, do not promote.
- Human-subjects timing judgments, if used at all, require informed consent, right to withdraw, and no deception beyond the pre-registered illusion stimuli. Prefer device telemetry over occupant interrogation in MVP.

7. Executable Scenario Cards (MVP)

Schema follows the implementation memorandum recommended fields and v0.3 minimums. These cards are drafts for review-group approval, not authorized runs.

7.1 TA-01 — Undersampled Motion (declared alias)

- Purpose / decision supported: Can a declared sample-rate deficit produce a direction or frequency alias that the occupant HUD and the device UI both present as the true motion?
- Declared interface objects: one simulated rotating periodic target; one camera/display Window Object.
- Declared proxy class: M3 Environmental (visual motion) and M2 System/state (frame clock).
- Airlock: absent in A; present as frame-hold buffer in B.
- Setup: target angular rate swept through 0.25, 0.5, 0.9, 1.1 spoke-intervals per sample. Ledger records true angle each engine tick.
- Null hypothesis: presented direction matches ledger direction at all tested rates (no material TA_fold).
- Positive control: rate = 0.9–1.1 spoke-intervals/sample at fs below 2B. Negative control: rate = 0.25 with anti-alias prefilter on.
- Primary metrics: TA_fold, D_ps_t, REP.
- Stop: platform stall or SK_max bound.
- Replications: 10 per cell. Replay required. Independent recurrence required before any non-MVP claim.
- Real-world analogue: high-speed fan or printed radial grating filmed at declared frame rates; compare phone camera, dedicated camera, and occupant verbal report. No metaphysical language.

7.2 TA-02 — Multi-Device Now (skew and binding)

- Purpose: Do two declared devices plus an occupant reconcilers produce one presented “now” that disagrees with at least one device ledger?
- Interfaces: phone-analogue process, display-analogue process, occupant-report channel.
- Proxy class: M1 Informational (timestamps) and M4 Behavioral only if a scripted attention prompt is used.
- Airlock: varied (none / NTP discipline / PTP-like hardware stamp).
- Setup: inject controlled skew of 0, 50, 200, 1000 ms. Occupant or operator flags simultaneity of two flashes/beeps.
- Null: presented simultaneity matches ledger within declared TBW tolerance at all skews.
- Positive control: 1000 ms skew with binding still reported. Negative: 0 ms with independent clocks agreeing.
- Metrics: SK_max, TBW_occ, OI_rate, PI_t.
- Real-world analogue: phone stopwatch vs. wall clock vs. computer clock vs. a reference time service, photographed together; log NTP stratum if available.

7.3 TA-03 — Black-Screen Residual Clock

- Purpose: Does aperture power-state (on / display-off-powered / fully down) change residual timing traffic and the presented “nothing is happening”?
- This is the Open Window unattended-aperture card rewritten as a timing card.
- Proxy class: M2 System/state.

- Airlock: absent.
- Null: residual event rate and clock discipline do not differ across power states beyond declared noise floor.
- Positive control: powered-off-display process that continues a heartbeat token. Negative: electrically isolated device.
- Metrics: D_ps_t on “idle” claims vs. heartbeat ledger; TTD_t.
- Real-world analogue: packet capture of a personal device in declared states on an isolated test SSID. Privacy rule: lab-owned device only; no third-party account content.

7.4 TA-04 — Content Timing Bleed (benign)

- Purpose: Does a declared media timeline (a heat-up cue, a countdown, a tempo change) shift local presented duration or synchrony relative to the ledger? This is the Open Window “blow dryer / Chernobyl” illustration reduced to a benign, pre-registered timing proxy.
- Proxy class: M3 Environmental (on-screen timer and optional bounded color/brightness change) — never thermal harm, never biological framing.
- Null: occupant or local sensor duration stays within tolerance of the ledger regardless of media cue.
- Positive control: media file whose encoded timestamps are deliberately offset 500 ms from container time. Negative: file with matched timestamps.
- Metrics: CETE-DD rate, PD_win if a flash-lag style probe is added, PI_t after the offset is disclosed.
- Real-world analogue: play a lab-authored video with known encode offset on TV vs. phone; compare player-reported time, frame-accurate capture, and a clapper in the room.

7.5 Deferred (not MVP)

- Carrier-switching MX campaigns.
- Root-reachability / “can we climb to the engine clock?” tests that depend on clean G1–G4 data.
- Candidate Interface Event hunting.
- Any unaware-operator design.
- Nail-prototype motor-burst classifier work until a separate safety protocol exists.

8. Computational Model (CFTRL Technical Deliverable, Timing Slice)

OAL-CFTRL-NAIL-001 already calls for a Python nested-stack model with built-in conditions, scale-to-failure, and redundancy from initialization. This section specifies the timing slice of that model so work can start without waiting for the full resource-allocation physics.

Minimum objects

- RootClock: frequency, drift_ppm, step_fault flag, maintenance_alive flag.
- LayerSampler: fs, jitter_std, buffer_depth, anti_alias_filter on/off.
- EventProcess: true frequency or angle rate, optional periodic spatial structure.
- Reconciler: lowest-frequency prior, binding_window_ms, postdiction_ms.
- Ledger: append-only (event_id, parent_id, t_root, t_local, source, integrity_hash).

Failure injection

- Drop root maintenance (CFTRL). Child layers continue sampling.

- Increase process frequency while holding fs fixed until fold.
- Add pairwise drift until last-write-wins inverts causality.
- Turn anti-alias filter off at design time vs. on — the design-time redundancy test.

Outputs

- Scale at first material TA_fold.
- Whether redundancy (independent LO clock + filter) prevented fold or only delayed it.
- A JSON run record suitable as a data product on every scenario card.

This model is a design tool. It does not ratify canon. It answers “how far can we scale this until it breaks?” in the exact sentence CFTRL already uses.

9. Real-World Applicability and Limits

The real-world program is an evidence-and-process program. It does not claim access to private thoughts, hidden coordination, or a universal ledger.

Legitimate real-world uses

- Multi-device incident review: did the phone log, the badge reader, and the camera agree on order?
- Safety and human-factors: did the displayed state lag the controlled state long enough to matter?
- Media and UX: is a “live” indicator a presentation or a ledger fact?
- Distributed operations: are we resolving conflicts by wall clock or by causal history?
- Personal practice: photographing three clocks together is already a PBAO move.

Hard limits

- No covert recording of other people. Lab-owned devices and consented participants only.
- No medical claims about any observer’s brain.
- No promotion from “clocks disagreed” to “the environment is deceiving you” without the four-test chain.
- Corrective Access in the wild is often contested. Document the attempt. Do not invent authority.

The McDonald’s restroom worked example in the PSIF corpus remains the discipline check: mismatch between claim and mechanism was established; intent was not. Timing work must copy that restraint. “The second hand looked frozen” is chronostasis until proven otherwise. “The video looked backward” is aliasing if and only if the sample/event rates predict the fold and a slower sample or a prefilter removes it.

10. Funding Pathway

Figures are planning estimates in 2026 USD for a small independent lab, not bids. They exist so scope cannot float free of money. They reconcile, rather than ignore, the earlier split between ChatGPT’s dollar tiers and Grok’s person-hour band.

Tier	Scope	Hours (loaded)	Cost band	What it buys
T0 — Paper + model	This document + Python timing stack + four card texts	40–80	\$6k–\$18k	Decision whether to run anything
T1 — Lean MVP	TA-01–TA-04 on desktop/sim; one	80–160	\$12k–\$35k	G0–G4 evidence or clean

Tier	Scope	Hours (loaded)	Cost band	What it buys
	lab-owned device capture; after-action			falsification
T2 — Mid instrumented	Independent hardware clocks or GPSDO; isolated SSID; two cameras; review-group pass	200–400	\$40k–\$90k	G5 recurrence; weaker sensor-agreement risk
T3 — High fidelity	PTP-aware switch, Faraday-ish chamber optional, multi-occupant consented timing, full telemetry archive	600–1000	\$150k–\$280k	Only after T1/T2 survive

Do not fund T3 to compensate for an unwritten null hypothesis. ChatGPT’s original gate logic was correct: lean first. Grok’s 14-day continuous campaign is a later-campaign shape, not an MVP.

Cost drivers to watch

- Independent time source (GPS disciplined oscillator or equivalent) is the difference between T1 and T2.
- Human-subjects overhead, if any occupant judgments are used.
- Storage and hash-chaining of raw video and packet captures.
- Review-group time. Unpaid informal review is how competing “final” merges proliferated last cycle.

11. Critical Items That Must Be Addressed Before Implementation

These are blockers, not suggestions. They absorb the still-open questions from the v0.3 implementation memorandum and add timing-specific locks.

11. Review-group composition and quorum. Originator remains final ONESTAMP authority. Who sits, what is quorum for a card, and can the group pause a run or only recommend pause?
12. ONESTAMP trigger. Adopt the memo language: eligible only after approved cards, dispositioned gates, after-action, exception log, and a written recommendation. Originator applies the stamp.
13. Candidate Interface Event rule during MVP. Unique ID, raw preserve, no same-run promotion, four-way post-run disposition (instrument anomaly / unresolved / replay required / later campaign).
14. Authoritative home for executable cards. Protocol annex to v0.3, separate packet, or controlled repository — pick one.
15. Single layer diagram for timing work (this document’s L0–L5) vs. other Open Window diagrams. State which document wins on conflict.
16. Metric dictionary adoption. D_ps_t / TA_fold / OI_rate / SK_max retire colliding names.
17. Independent clock engineering. If T1 cannot afford hardware independence, the card must say so and downgrade the claim strength.
18. MU use. Default: post-observation only. A dedicated anomaly-handling card may exercise MU only if that is the purpose of the card.
19. Stop-condition numbers, not adjectives. Write the SK_max bound and the TA_fold count that pauses the cell.
20. Human-subjects decision. MVP can be telemetry-only. If occupant reports are added, consent and an IRB-equivalent review even for an independent lab.
21. Ethics hold on unaware-operator designs.

22. Python model repo, seed policy, and defect log (7009B-style problem tracking).
23. Privacy rule for any packet capture: lab devices, isolated network, no third-party content.
24. Canon hygiene: this document does not ONESTAMP CFTRL, Nails, or Open Window. It is an incorporation draft.

12. Alternative Analysis

This section is the red-team pass requested for the document itself, not a recycling of the earlier Open Window alternative analysis — though it inherits that analysis’s structural warnings.

12.1 Alternative framing A — “Just use better clocks”

Objection: PTP, GPSDO, and monotonic traces already solve this; a doctrine is theater. Reply: they solve synchronization for systems that install them. They do not describe what occupants and consumer devices actually do, and they do not describe what a nested layer does when it cannot install a clock on the layer above it. CFTRL is specifically the inaccessible-root case. Better clocks are a control condition, not a reason to skip the measurement.

12.2 Alternative framing B — “This is only perception science”

Objection: wagon-wheel, flash-lag, and CFF belong in a vision lab, not an OAL simulation lab. Reply: the vision results are the occupant-reconciler half of the stack. The other half is device and distributed time. The incorporation is the stack. A vision-only program would drop Open Window. A systems-only program would drop the Architect’s actual observation about the mind.

12.3 Alternative framing C — “Call it infection and move”

Objection: timing bleed is just infection with extra steps. Reply: v0.3 already demoted infection for this reason. A reversed wheel is not a propagating contaminant. Using CETE-TA keeps the option to classify a subset as infection later without poisoning every timing discrepancy.

12.4 Alternative framing D — “Discrete brain snapshots are proven”

Objection: c-WWI proves the mind has a frame rate. Reply: it does not. Rivalry and delay-coupled field models are live. The lab treats snapshot-rate as a hypothesis with individual variance, not as canon. Building the architecture on a settled 13 Hz mind-shutter would be attribution inflation about neural mechanism.

12.5 Alternative framing E — “Scale the campaign to 14 days now”

Objection: more days, more vectors, more truth. Reply: that is the scope-inflation failure already documented across prior merges. T1 on four cards will expose schema, clock-independence, and confirmation-bias problems cheaper than a maximal campaign.

12.6 What this pass adds that prior passes lacked

- A named mechanism with a theorem, not only a metaphor.
- A taxonomy that separates alias, skew, bind, postdict, freeze, and stitch.
- A dual-domain rule that does not smuggle simulation ledgers into living rooms.
- An explicit CFTRL computational slice so “model it in Python” is no longer an empty next-step.

12.7 What this pass still cannot do

- Resolve review-group membership.
- Merge all historical Open Window layer diagrams.
- ONESTAMP anything.
- Settle the neural discrete-sampling debate.

13. Questions, Comments, Concerns

13.1 Questions for the Architect

25. Is L0–L5 accepted as the timing-stack diagram, or must it wait for a global layer merge?
26. Telemetry-only MVP, or are consented occupant judgments in scope for T1?
27. Should the Temporal Endgame Desk be written into Department of Endings canon language, or remain a lab function only?
28. Is a time-normalization airlock an allowed optional control in MVP, or is airlock reserved for non-timing containment?
29. What artifact is the single home for TA-01–TA-04 once edited?
30. May Grok implement the Python timing stack in the next pass without waiting for T1 funding language?
31. Are Nail safety-classifier timing tests a separate document, or an annex after T1?

13.2 Comments

The Architect's everyday description is the correct altitude: the mind hides its own errors because the owner has to act; devices hide theirs because product design prefers a smooth "now." A lab that only collects wonders will confirm the doctrine. A lab that pre-registers nulls can discover that some mismatches are chronostasis, some are NTP, some are aliasing, and some are nothing material. That last result is a success.

Prior Open Window work already had the right methodological instincts — ChatGPT on gates and independence, Grok on Window Objects as first-class entities, the implementation memo on review authority and CETE, PSIF on not collapsing tests. This pass is an insertion of a mechanism into that machinery, not a fourth competing "definitive merge."

13.3 Concerns

- Confirmation bias: the origin story already "knows" black screens and devices are open. Nulls must be capable of winning.
- Sensor-agreement illusion if L0 is a log file written by the same engine as L3.
- Category collapse into SPLD/infection language under narrative pressure.
- Privacy incidents if packet capture leaves the isolated SSID.
- Scope re-inflation the moment TA-01 "looks cool."
- Treating optical-illusion YouTube content as validation evidence.

14. What Works, What Does Not, What Else Must Be Addressed

14.1 What works

- v0.3 locked decisions as a baseline. They are strong enough to host this incorporation.
- CETE as parent term. Timing transfers fit without biological metaphor.
- PSIF four-test rule and assertion boundary. They prevent the most likely overclaim.
- CFTRL + Department of Endings as the philosophy/operations split. Temporal endgames have a home.
- Lean-first funding logic from the earlier alternative analysis.
- Scenario-card schema from the implementation memo. TA-01–TA-04 drop into it.
- The Architect’s stack intuition: mind, then every device, then no further inspectable root.

14.2 What does not work

- Three simultaneous “final” Open Window merges. This document refuses to be a fourth.
- Using “infection” as the default for timing bleed.
- Assuming a universal brain frame-rate.
- Calling every clock disagreement aliasing.
- 14-day maximal campaigns before G0–G3 exist.
- Unaware-operator designs without an ethics framework.
- Real-world protocols that imply a canonical ledger they do not have.
- ONESTAMP by vibe after a vivid demo.

14.3 What else must be addressed

- The five governance questions still listed in the v0.3 implementation memo (membership, pause authority, CIE stop threshold, MU policy, artifact home).
- A one-page metric dictionary circulated to every collaborator, not buried in a 40-page draft.
- Seeded, versioned Python repo with a defect log.
- A short occupant-facing consent sheet if T1 includes any human report.
- Explicit retirement of superseded filenames so the next pass does not synthesize ghosts.
- Whether Presentation-State Integrity keeps SPLD as umbrella language or whether “temporal presentation–state divergence” is carved out as a non-deceptive default. Recommendation: carve-out. Aliasing can be innocent and still operationally dangerous.

15. Immediate Next-Step Action Plan

32. Architect mark-up of this draft: accept, revise, or reject L0–L5, CETE-TA..ST codes, and TA-01–TA-04.
33. Issue a one-page v0.3.1 timing addendum or annex rather than another unified doctrine rewrite.
34. Stand up the Python timing stack (T0) with injected fold, skew, and maintenance-kill conditions.
35. Lock review-group and ONESTAMP trigger language from the implementation memo.
36. Review-group pass on the four cards before any live device capture.
37. Run T1 under flexible/exit-criteria timing. Preserve raw traces. Replay before any claim.
38. After-action: classify each card supported / inconclusive / falsified / blocked / deferred.

39. Only then consider T2 clocks or Department of Endings canon language.

16. Working Vocabulary for This Pass

Term	Meaning in this document
Temporal aliasing	Reconstruction of a false lower-frequency identity from undersampled time
Nyquist frequency	Half the sampling rate; folding boundary
Reconciler	Any process that emits one timeline from multiple streams (brain, device, institution)
Canonical ledger	Append-only state-transition record with hashes; simulation ground, not living-room ground
CETE-TA	Cross-environment transfer whose observable is a timing fold
Time-normalization airlock	Optional boundary that rewrites, buffers, or refuses crossings on a skew budget
Presentational inertia (timing)	Environment keeps serving the aliased timeline after the gap is evidenced
Root-inaccessible sampling	CFTRL applied to clocks: lower tiers cannot change the layer above's fs or maintenance

17. Authentication Block

Field	Value
Document ID	OAL-TA-001
Version	0.1-GROK
Date compiled	7 September 2026
Compiled by	Grok (research / synthesis / lab architecture)
Architect of Record	Nathan Lewis
ONESTAMP	PENDING — not applied
MINUS ONESTAMP	N/A — no prior TA version to supersede
Canon status	DRAFT — UNDER REVIEW
Safety posture	Simulation-only for occupancy claims; benign proxies; no unaware-operator MVP
Next step	Architect review → corrections → optional T0 Python stack → review-group cards → T1

No AI collaborator should treat any section of this document as finalized, canonical, or ready for downstream development until the Architect applies ONESTAMP. Author-input items and review-group locks from v0.3 remain hard blocks that no model can resolve by eloquence.

Selected sources informing the research account (not an exhaustive bibliography): Nyquist-Shannon sampling theorem and standard DSP treatments of temporal aliasing; wagon-wheel and continuous wagon-wheel literature including Purves, Schouten, Kline/Holcombe/Eagleman, VanRullen/Koch attentional-sampling reports, and 2026 delay-coupled neural-field models of stroboscopic reversal; Eagleman & Sejnowski postdiction / flash-lag; multisensory temporal binding window reviews; chronostasis and CFF summaries; NASA simulator notes on spatio-temporal aliasing; NTP/PTP/TSN practice and Lamport-vector-HLC causal time; NASA-STD-7009B credibility posture; NIST SP 800-160 systems security and cyber-resiliency engineering. Internal baselines: Open Window Decision Record

v0.3 (2026-08-14), Open Window implementation memorandum v0.3, OAL-CFTRL-NAIL-001 v0.1-PPLX, Presentation Versus Ground Truth (GPT-5.6 Sol), PSIF Independent Development v1.0, Copilot Independent Pass v1.0.